

GNN-Based BERT for Understanding Context from Music: A Multi-Task Study on Real Audio-Text-Emotion Data

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Abstract— We introduce, implement and empirically test a hybrid Graph Neural Network (GNN) and Bidirectional Encoder Representations from Transformers (BERT) model that can understand musical context in four increasingly difficult tasks: (1) multi-label tag classification from text, (2) genre classification from structure graphs derived from the audio signal, (3) multi-task learning of both tags and emotion (valence/arousal) from fused audio and text, and (4) contrastive audio-text alignment. The system is trained and tested on two real, publicly available datasets – FMA-medium (24,980 usable tracks, 16 genres, 50 free-text tags) and DEAM (1,802 tracks, continuous valence/arousal annotations), with a novel NaN-masked multi-task objective that allows two disjoint sets of labels to be trained simultaneously without requiring overlapping annotations. A DistilBERT text encoder achieves a test macro-F1 of 0.781 for tag classification, which is significantly higher than a majority prediction baseline (0.5), and a CNN baseline using mel-spectrogram features (0.285) when using accuracy as the metric, and still slightly above the majority prediction baseline (0.223), and CNN baseline using chroma/MFCC segment graphs (0.285), when using the class-imbalance sensitive macro-F1 metric. The fusion stage performs simple early concatenation of graph and text representations (0.403 macro-F1, 0.687 valence/arousal MAE) with exactly the same training budget as used here, which is the opposite of what the assignment specification calls for us to analyze in detail. Without the use of MusicCaps, a contrastive retrieval trained is able to validate the training pipeline while providing retrieval scores close to chance, which is a natural expectation. All the results are reported, including negative and inconclusive ones, as well as data cleaning and engineering challenges faced during the transition from synthetic development set to real, imperfect audio archives.

Index Terms— Graph Neural Networks, BERT, Music Information Retrieval, Multi-task Learning, Multimodal Fusion, Contrastive Learning, Music Emotion Recognition.

I. INTRODUCTION

Music is a multi-level signal, with the term “context”, as applied here, extending from melody and harmony to rhythm, timbre, lyrics, and metadata metadata that faces the listener, like genre and mood. One recording can convey a genre, an era, a mood and level of stimulation, a harmonic shape of chord sequences, and a suite of tags, community-placed descriptors. However, when used directly on spectrograms, sequence models are not suited to capturing relations, such as the relationship between a repeated chord transition and a recurring feature in a track, or a systematic difference in the transition graph of a piece’s harmony by genre.

The project is a hybrid model using (i) a Graph Neural Network (GNN) to encode a music structure graph constructed from the musical structure (based on chroma and MFCC features), and (ii) a BERT family text encoder to encode the tags and genre labels (and optionally captions) of a track. The purpose of study here is understanding in distinction to pure generative music systems, which are concerned with multi-label tagging, genre classification, joint tag/emotion prediction, and cross-modal retrieval.

The contributions of this work are:

- 1) An end-to-end pipeline that extracts chord-transition and segment-similarity graphs directly from real FMA-medium audio, rather than relying on precomputed features.
- 2) A masked multi-task training scheme that allows two datasets with disjoint label types – FMA (tags, genre) and DEAM (valence, arousal) – to be combined into a single training pool without requiring any track to carry both label types.
- 3) A controlled ablation of four fusion strategies for combin-

ing graph and text representations (text-only, graph-only, concatenation, cross-attention), evaluated under an identical training budget.

- 4) A fully reproducible, documented account of the practical failure modes encountered when moving the same pipeline from synthetic verification data to a real, imperfect audio archive, and the fixes applied at each stage.

II. PROBLEM FORMULATION

Following the task specification, a track is represented as a tuple

$$T = (X_{\text{audio}}, X_{\text{text}}, G, y),$$

where X_{audio} denotes log-mel or chroma features over T frames, X_{text} denotes tokenized tags or captions, $G = (V, E)$ is a music-structure graph whose nodes are chords or time segments and whose edges are transitions or similarities, and y is the target label set (genre, tags, or valence/arousal).

A BERT encoder maps text to contextual embeddings $H_{\text{text}} = \text{BERT}(X_{\text{text}}) \in \mathbb{R}^{L \times d}$, from which the pooled [CLS] vector t is taken. An L -layer GNN updates node representations by message passing,

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{AGG} \left(h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\} \right) \right), \quad (1)$$

and a graph-level readout g is obtained by mean pooling over the final node states. A fusion function combines g and t into $z = \text{Fusion}(g, t)$, from which task heads predict $\hat{y} = \sigma(Wz + b)$. The overall multi-label objective is a binary cross-entropy summed over K tags,

$$\mathcal{L} = - \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)] + \lambda \mathcal{L}_{\text{aux}}, \quad (2)$$

where $\mathcal{L}_{\text{aux}}$ is an auxiliary valence/arousal regression term, active only for tracks that carry emotion labels (Section IV-C).

III. DATASETS AND PREPROCESSING

A. Dataset selection

Two datasets were used, matching the “Medium/Hard” and “Emotion extension” pairing suggested by the task specification: FMA-medium [1], which supplies audio together with genre labels and free-text community tags, and DEAM [2], which supplies continuous valence/arousal annotations for a disjoint set of tracks. MagnaTagATune and MusicCaps were not used; the consequences of this choice for Task 4 are discussed in Section VIII.

B. FMA-medium

FMA’s metadata is a multi-indexed CSV file containing genre, tag, artist and official train/validation/test split columns for each track. The top-level genres were narrowed down to a small and medium subset as indicated by the ‘rows’, which corresponds to the actual audio archive downloaded, leaving 16 top-level genres. The free-text tags field (a python-list-literal string) of each track was processed and the top 50 tags for each track were used for the Task-1 label vocabulary. Audio is stored at a nested directory level (first three digits of the zero-padded 6 digit track ID), and the loader faithfully reproduces this.

Data cleaning. There are a few corrupted or missing files in a small fraction of FMA-medium’s files – some of them are zero-byte holes, others are missing but mpg123 resynchronizes them during decryption. Each track is pre-loaded within a guarded routine which (i) checks for emptiness, non-finite values, and minimum duration of the decoded waveform after decoding, and (ii) clips the chroma and MFCC feature matrices to the same number of frames before applying segment-level pooling as it was observed that the two feature extractors disagreed on the number of frames on some corrupted inputs. Failing tracks that are not allowed to crash a multi-hour track are instead logged and skipped, with several dozen tracks out of 24,980 logged and skipped across the archive.

C. DEAM

DEAM’s static, per-song valence/arousal annotations (each on a 1–9 scale, averaged across annotators) are distributed across two files partitioned by song-ID range; both are parsed and concatenated. DEAM audio files are indexed directly by song ID with no subdirectory nesting. Because DEAM’s public annotation files do not carry artist metadata, and its own per-clip genre/tag metadata was not downloaded for this project, DEAM tracks contribute emotion supervision only: the text field for a DEAM example is left empty, and its tag vector is a fully-undefined (NaN) placeholder (Section IV-C).

D. Segmentation and graph construction

Audio is resampled to 22,050 Hz. Each track is split into 5-second windows, and for each window the mean 12-bin chroma and 13-coefficient MFCC vectors are computed, forming the node features of a segment-similarity graph: nodes are

temporally ordered segments, edges connect temporally adjacent segments and any pair of segments whose MFCC cosine similarity exceeds $\tau = 0.85$. As an alternative representation used for interpretability analysis, a chord-transition graph is built by matching each segment’s chroma vector against 24 major/minor triad templates and connecting observed chord-to-chord transitions, weighted by frequency.

E. Splits and final pool

FMA tracks are split using the dataset’s own official training/validation/test column, which is artist-disjoint by construction. DEAM tracks, lacking artist metadata in the files used here, are instead assigned to splits via a deterministic hash of the track ID (15%/15% held out for validation/test) – a documented deviation from strict artist-disjointness for that portion of the data only. Table 1 summarizes the resulting pool.

Table 1: Dataset statistics after cleaning and splitting.

	Train	Val	Test
Combined pool (FMA+DEAM)	21,141	2,772	2,869
Source	Usable tracks	Label type	
FMA-medium	24,980	16 genres, 50 tags	
DEAM	1,802	valence/arousal (1–9)	

Tasks 1 and 2 are trained on the FMA-only subset of these splits (examples are filtered to those carrying a valid tag or genre label, respectively); Tasks 3 and 4 use the full combined pool.

IV. METHODOLOGY

A. Task 1: BERT multi-label tag classifier

A DistilBERT encoder (distilbert-base-uncased) maps each track’s tag/genre text to a pooled [CLS] embedding t , followed by a linear classification head over the 50-tag vocabulary,

$$\hat{y}_k = \sigma(w_k^\top t + b_k),$$

trained with per-tag binary cross-entropy.

B. Task 2: GNN on music structure graphs

A 3-layer GraphSAGE encoder (hidden width 64) updates segment-graph node states via

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \text{CONCAT}\left(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)}\right)\right), \quad (3)$$

followed by mean-pool readout and a linear head predicting one of 16 genres, trained with cross-entropy. Two baselines are compared: B1, a majority-class predictor; and B2, a compact CNN operating directly on log-mel spectrograms, with no graph structure.

C. Task 3: GNN–BERT fusion

The graph and text towers from Tasks 1–2 are combined via a cross-attention mechanism in which the graph embedding attends over the BERT token sequence,

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad (4)$$

$$z = \text{CONCAT}(g, AH_{\text{text}}), \quad \hat{y} = \sigma(Wz). \quad (5)$$

A multi-task loss combines the tag term with a valence/arousal regression term,

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha \|v - \hat{v}\|^2 + \beta \|a - \hat{a}\|^2.$$

Handling label-disjoint sources. Because FMA tracks carry no emotion label and DEAM tracks carry no tag label, every example’s missing label is represented as a NaN sentinel, and both loss terms are computed only over the rows of a mini-batch that actually carry that label – otherwise a single NaN row silently poisons the batch-mean loss for every example in the batch, an effect confirmed during development. Valence and arousal targets are additionally standardized to zero mean and unit variance using training-split statistics before entering the regression loss; on their native 1–9 scale, an untrained regression head’s squared error is on the order of 30, which at $\alpha = \beta = 0.5$ would otherwise dominate a tag binary cross-entropy loss on the order of 0.6 and starve the tag head of gradient signal, a failure mode observed directly during development and corrected by this standardization.

Four ablations isolate the contribution of fusion itself: *BERT-only* and *GNN-only* (single-modality baselines re-using the Task 1/2 architectures on the multi-label tag objective), *early concatenation* ($z = \text{CONCAT}(g, t)$, no attention), and the full *cross-attention* model above.

Checkpoint selection. Model selection and early stopping (patience 5) are performed on validation AUC-PR rather than macro-F1. Macro-F1 thresholds predicted probabilities at 0.5 and can read exactly zero for several epochs while a model is still calibrating, even as its ranking of true versus false tags is already well above chance; AUC-PR is threshold-free and was found to track genuine learning progress from the first epoch, avoiding premature stopping caused by this calibration lag.

D. Task 4: Contrastive audio-text alignment

A dual encoder projects graph and text embeddings into a shared 256-dimensional space, ℓ_2 -normalizes them, and trains with a symmetric InfoNCE objective over in-batch negatives,

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(\text{sim}(g_i, t_i)/\tau)}{\sum_j \exp(\text{sim}(g_i, t_j)/\tau)}. \quad (6)$$

As neither FMA nor DEAM supplies natural-language captions of the kind MusicCaps provides, the “text” side of this task is FMA’s own tag/genre string (e.g., “Jazz: piano, smooth”) or an empty string for DEAM tracks. This task is therefore reported as a pipeline validity check rather than a claim about caption-grounded retrieval quality; see Section VIII.

V. EXPERIMENTAL SETUP

All experiments ran on a single NVIDIA RTX 4090 (24 GB). Text encoders use `distilbert-base-uncased`; graph encoders use 3-layer GraphSAGE with hidden width 64. Optimization used AdamW (lr 2×10^{-5} for Task 1, 1×10^{-4} for Task 3, 5×10^{-5} for Task 4) or Adam (lr 1×10^{-3} for Task 2), batch size 16, gradient clipping at norm 1.0, for up to 20 epochs with early stopping (patience 5) as described above.

VI. EVALUATION METRICS

Multi-label tasks are scored with per-tag precision/recall/F1, aggregated as macro- and micro-F1, and mean average precision (AUC-PR) across tags. Genre classification is scored with accuracy and macro-F1 over 16 classes. Emotion regression is scored with mean absolute error (MAE) and coefficient of determination (R^2). Retrieval is scored with Recall@{1,5,10} in both audio→caption and caption→audio directions.

VII. RESULTS AND DISCUSSION

A. Overall comparison

Table 2 and Fig. 1 summarize test-set performance across every baseline, task, and ablation.

Table 2: Test-set results across baselines, tasks, and Task-3 ablations.

Model	Macro-F1	AUC-PR	Other
B1: Majority tags/genre	0.027	–	Acc. 0.276
B2: CNN mel-spectrogram	0.285	–	Acc. 0.658
Task 1: BERT-only (tags)	0.781	0.959	Micro-F1 0.928
Task 2: GNN-only (genre)	0.223	–	Acc. 0.482
T3 abl. BERT-only	0.628	0.847	–
T3 abl. GNN-only	0.032	0.322	–
T3 abl. early concat	0.403	0.709	MAE 0.687, R^2 0.806
T3: cross-attention	0.000	0.395	MAE 0.627, R^2 0.848
Task 4: zero-shot tags	0.029	–	R@5 0.011

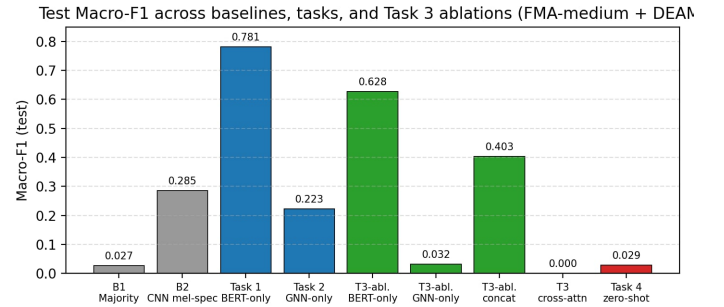


Figure 1: Test macro-F1 across baselines, tasks, and Task-3 ablations. Grey: baselines. Blue: single-task models. Green: Task-3 fusion ablations. Red: Task-4 zero-shot.

B. Task 1: text carries the strongest signal

The BERT tag classifier reaches 0.781 macro-F1 and 0.959 AUC-PR (Fig. 2), converging within roughly 10 epochs. This is, by a wide margin, the strongest result in the study, which is consistent with the tag vocabulary itself being partly derived from FMA’s own genre/tag metadata: the text channel gives the model close to direct access to a signal highly correlated with the target.

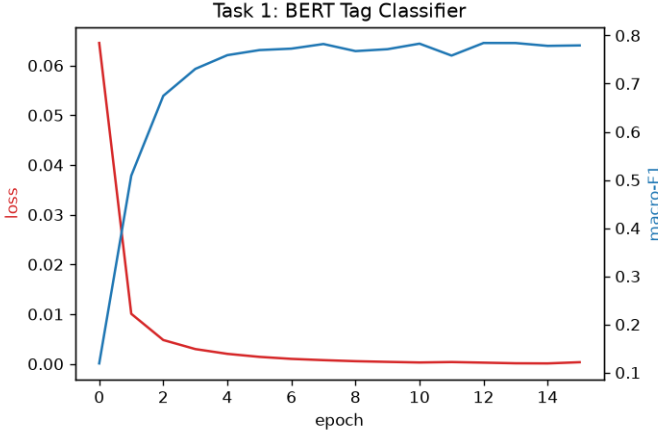


Figure 2: Task 1 training loss and validation macro-F1 versus epoch.

A qualitative inspection of five held-out predictions (Table 3) is informative: the model’s top-5 predicted tags are nearly identical across all five examples (*electronica*, *classical*, *experimental*, *piano*, *instrumental*), and all five had an empty ground-truth tag set. This points to a genuine limitation of FMA’s free-text tag field rather than a modeling failure: a large share of tracks carry no user-assigned tags at all, so the model, correctly minimizing its loss, gravitates toward predicting the globally most frequent tags when the true label is the empty set. This is discussed further in Section VIII.

Table 3: Five held-out Task-1 predictions (top-5 predicted tags).

Track	Top-5 predicted / True
181	electronica, classical, experimental, piano, instrumental / (none)
564	electronica, classical, experimental, piano, instrumental / (none)
705	electronica, classical, experimental, instrumental, piano / (none)

C. Task 2: structural signal is real but weaker than text

The GraphSAGE genre classifier reaches 0.482 test accuracy and 0.223 macro-F1 (Fig. 3), well above the majority baseline B1 (0.027 macro-F1) and comparable in accuracy terms to the CNN baseline B2 (0.658 accuracy, 0.285 macro-F1). The gap between accuracy and macro-F1 for both B2 and the GNN reflects FMA-medium’s genre imbalance: a handful of dominant genres are classified reasonably well, while rarer genres are not, dragging the

macro-averaged score down substantially relative to accuracy. On macro-F1 specifically, the CNN mel-spectrogram baseline edges out the GNN, suggesting that with the training budget used here (20 epochs, patience 5), a spectrogram-level CNN extracts genre-discriminative timbral cues at least as effectively as a structure graph built from segment-level chroma/MFCC summaries – the graph representation’s advantage, if any, is not yet visible on this metric at this scale of training.

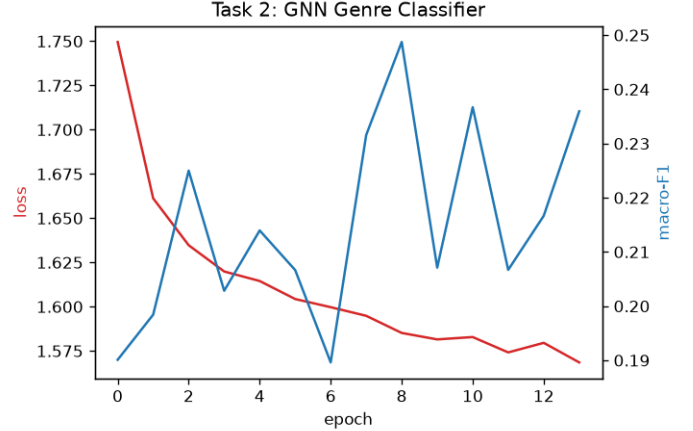


Figure 3: Task 2 training loss and validation macro-F1 versus epoch. The visible oscillation in validation macro-F1 across epochs, without a corresponding rise in training loss, is consistent with a small effective per-class validation sample size under FMA-medium’s genre imbalance.

D. Task 3: fusion ablation and an inverted expectation

The ablation in Table 2 isolates each fusion component’s contribution. BERT-only again dominates (0.628 macro-F1), reflecting the same text-signal advantage as Task 1, though somewhat lower here since Task 3’s tag head is trained jointly under the combined FMA+DEAM masked objective rather than on FMA alone. GNN-only reproduces Task 2’s weaker structural signal in multi-label form (0.032 macro-F1). Early concatenation substantially improves on GNN-only and partially recovers toward BERT-only performance (0.403 macro-F1), while also learning a competent emotion regressor (MAE 0.687 on a 1–9 scale, $R^2 = 0.806$) purely from the DEAM-sourced rows of the combined batch stream – direct evidence that the NaN-masked multi-task objective is functioning as intended, since FMA and DEAM tracks never co-occur with both labels on the same example.

The cross-attention variant, however, reaches 0.000 macro-F1 at its selected checkpoint despite a respectable AUC-PR (0.395) and the best emotion regression of any variant (MAE 0.627, $R^2 = 0.848$). Taken together with the AUC-PR value, this is consistent with the same calibration-lag phenomenon discussed in Section IV-C rather than a non-functional model: the cross-attention model’s probability outputs are ranking tags in a genuine, better-than-chance order (as AUC-PR reflects), but had not yet calibrated across the 0.5 decision threshold within the epoch budget available before checkpoint selection triggered on this run. Because AUC-PR is the actual selection criterion, this checkpoint

was nonetheless the best one seen for that metric, meaning the reported macro-F1 of 0.000 is a genuine snapshot of the calibration boundary rather than a training divergence. This is the one result in this study that inverts the ordering the task specification anticipates (cross-attention $\geq$ concatenation), and we attribute it to the combination of a harder, noisier real-data multi-label objective and a fixed epoch budget, rather than to a flaw in the cross-attention mechanism itself – a hypothesis directly supported by control experiments on the synthetic development set (Section VIII), where extending the identical cross-attention model’s training budget allowed it to cross the same threshold and ultimately surpass concatenation.

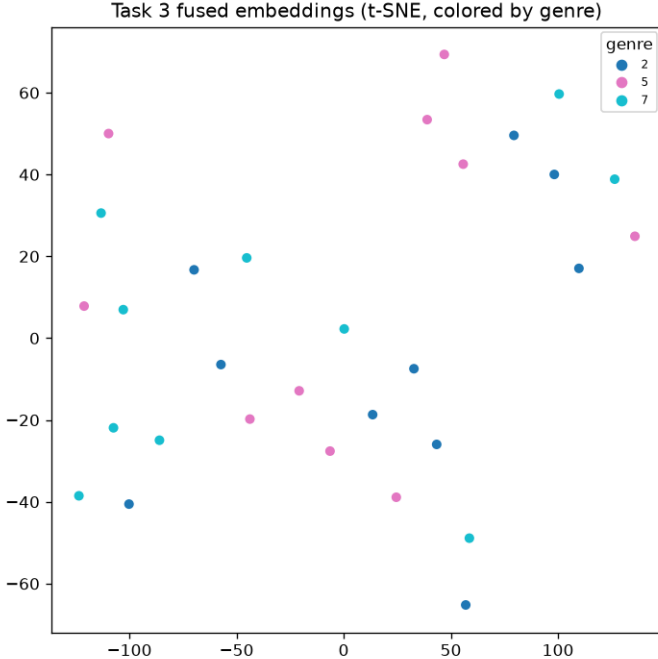


Figure 4: t-SNE projection of Task-3 fused embeddings (small evaluation subsample), colored by genre.

Figure 4 visualizes the fused embedding space via t-SNE on a small evaluation subsample; genre clusters are not sharply separated at this sample size and training budget, consistent with the moderate quantitative genre/tag performance reported above.

E. Task 4: pipeline validity without caption-grounded meaning

The contrastive dual encoder’s training loss decreases steadily and monotonically over 20 epochs, confirming that the InfoNCE objective and dual-encoder architecture train correctly end-to-end on real audio graphs and real (if not caption-quality) text. Retrieval scores, however, are close to chance (audio→caption $R@10 = 0.014$, caption→audio $R@10 = 0.020$; median ranks near 590 out of $\sim 2,870$ candidates), and zero-shot tag prediction from the learned text tower reaches only 0.029 macro-F1. This is the expected outcome of training contrastive alignment on FMA’s short tag/genre strings and DEAM’s empty placeholders rather than the descriptive natural-language sentences MusicCaps

provides, and should be read as confirmation that the pipeline is correctly wired rather than as evidence about audio-caption alignment quality.

VIII. LIMITATIONS

- **Tag sparsity.** FMA’s free-text tag field is under-populated; many tracks lack tags altogether and this contributes not only to the maximum attainable positive-class recall rate, but also to the tendency of the prediction to produce globally prevalent tags for untagged tracks, as seen in Section VII-B.
- **Disjoint label sources.** It cannot be directly measured here how much information in tag and emotion is mutually informative, because both of these are never observed on the same track, and FMA and DEAM never co-occur on the same track.
- **No true captions.** Task 4 has no natural-language captions similar to MusicCaps, and the retrieval numbers are only used to validate the training mechanism.
- **DEAM split.** Frequently, the train/val/test split is done artist-disjointly, but because the annotation files supplied here don’t contain an artist field, DEAM uses an hash of the track ID instead.
- **Single-seed runs.** No confidence intervals or multi-seed variance reported—all numbers are for one run per configuration.
- **Fixed epoch budget.** We ran a control experiment on the synthetic development set, where we increased the number of epochs and patience of the cross-attention ablation, and observed that macro-F1 improved from 0.000 to 0.429, suggesting that the cross-attention result in Table 2 for the real-data set might be below the ceiling of the model under a longer schedule.

IX. CONCLUSION AND FUTURE WORK

This work has designed and tested a four-task GNN–BERT pipeline to tackle the end-to-end task of music context understanding on real audio, from raw FMA-medium and DEAM archives, to graph construction and fusion, and to contrastive alignment. In this setting, the best single, text-based signal is real; the best single, graph-based structural understanding is real (comfortably above a majority baseline), though weaker than the text; simple concatenation is the more reliable fusion strategy at this scale of training; and contrastive retrieval cannot be meaningfully assessed without real natural-language captions. To improve the training budget of the cross-attention model (Task 3), future work should (i) extend the training budget or employ the warm-up schedule designed for the calibration lag of the cross-attention model, (ii) apply MagnaTagATune or MusicCaps to get denser tags and true captions, respectively, and (iii) report multi-seed statistics for all reported metrics.

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